

01BWAY01

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Problem

A set of 8 problems was prepared for an examination. Each student was given 3 of them. No two students received more than one common problem. What is the largest possible number of students?

Solution. Consider a matrix where the columns represent the problems and the rows represent the students. Say there are n students, so there are n rows.

We will label the element $a_{ij} = 1$ if the i th student has received the j th problem and 0 otherwise. So we have $3n$ 1s.

We begin with the crucial claim:

Claim 1 — Each problem can be received by at most 3 students.

Proof. For the sake of contradiction, say at least 4 people have received the same problem. As a result, there is some column with ≥ 4 1s. The rows of these 1s must have 2 other 1s as well. Since we have ≥ 8 1s left but only 7 columns left, due to PHP, there is some column which again has 2 1s from the same rows. This contradicts the fact that no students received more than one problem. $\square$

Since we have 8 columns and each has at most 3 1s, there can be at most 24 1s. So we have $3n \leq 24 \implies n \leq 8$. It is enough to show that $n = 8$ does work.

Observe the following matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$